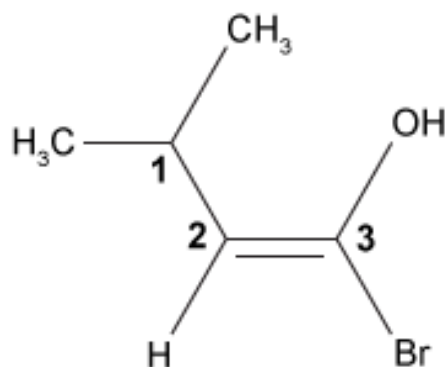


Researchers want to synthesize valine from isovaleric acid. The enol shown below, with certain carbon atoms labeled, is formed from the reaction of isovaleric acid with PBr_3 .



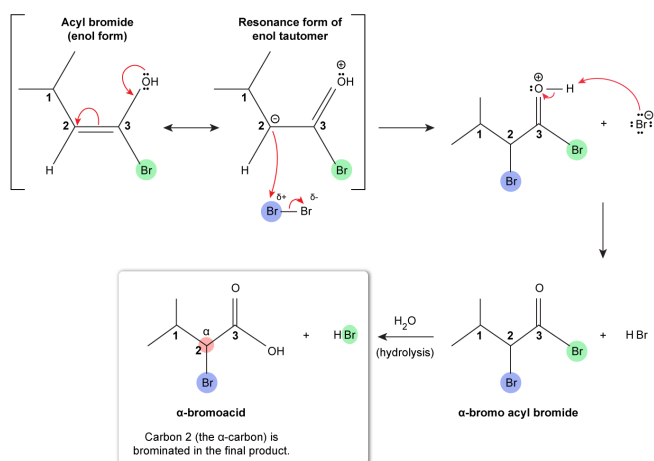
If Br_2 and then water are added to the enol, which of the following carbon atoms will be brominated in the final product?

- ☐ A. Carbons 1 and 2 only [10%]
- ✓ ☒ B. Carbon 2 only [43%]
- ☐ C. Carbons 1 and 3 only [10%]
- ☐ D. Carbon 3 only [36%]

Correct

43% answered correctly

Explanation:



Reactions at the α -position (ie, 2-position) of carboxylic acids are an extremely important class of reactions in organic chemistry and biochemistry.

Substitution of bromine on the **α -carbon** of a **carboxylic acid**, known as the Hell–Volhard–Zelinsky (HVZ) reaction, begins with the **addition of PBr_3** . This

step converts the carboxylic acid to an acyl bromide, which can then **tautomerize** from the keto form to the enol form.

When Br_2 is added to the reaction, the α -carbon of the enol acts as a nucleophile and attacks bromine. One of the bromine atoms in Br_2 withdraws electrons away from the other, giving it a partial positive charge and making it a good electrophile. The **electrophilic bromine is added to the α -carbon**, forming an α -bromo acyl bromide and HBr .

When water is added, it hydrolyzes the acyl bromide by attacking the electrophilic **carbonyl** carbon while the **bromine on the carbonyl carbon acts as the leaving group**. This yields an **α -bromoacid**, which is a useful intermediate in the **synthesis of α -amino acids** such as valine.

For these reactions, the carbon adjacent to the carbonyl group (carbon 2) is the α -carbon. The α -carbons in enols are susceptible to substitution because the carbocation intermediate formed during the substitution reaction is stabilized by **resonance** with the enol hydroxyl group. In this case, the α -carbon becomes brominated after the addition of Br_2 . Note that a **halonium/bromonium intermediate** is *not* formed from the reaction of the alkene $\text{C}=\text{C}$ bond with bromine because the electron-withdrawing O and Br groups destabilize any positive charge localization on the carbonyl carbon. The acyl bromide bromine on the carbonyl carbon (carbon 3) is **eliminated during hydrolysis**.

Therefore, **carbon 2** is the only carbon to remain brominated after hydrolysis, yielding the final product, an α -bromoacid.

(Choices A, C, and D) Although carbon 2 is brominated in the final product, carbon 1 does not participate in the reaction. The bromine on carbon 3 is eliminated during hydrolysis when water is added.

Educational objective:

The α -carbon of a carboxylic acid is the carbon atom adjacent to the carbonyl carbon, and it is susceptible to substitution reactions such as bromination. The carbonyl carbon of a carboxylic acid may be temporarily brominated, but this carbon is susceptible to hydrolysis when water is added.